

Artificial Intelligence Foundation – JC3001

Lecture 36: Section 3 Review

Prof. Aladdin Ayesh (aladdin.ayesh@abdn.ac.uk)

Dr. Binod Bhattarai (binod.bhattarai@abdn.ac.uk)

Dr. Gideon Ogunniye, (g.ogunniye@abdn.ac.uk)

October 2025

Material adapted from:

Russell and Norvig (AIMA Book): Chapter 5

Russell and Norvig (AIMA Book): Chapter 17/18 (17.1/18.2)

Shoham and Leyton-Brown (Game Theory)

Course Progression

- Part 1: Introduction
 - ① Introduction to AI ✓
 - ② Agents ✓
- Part 2: Problem-solving
 - ① Search 1: Uninformed Search ✓
 - ② Search 2: Heuristic Search ✓
 - ③ Search 3: Local Search ✓
 - ④ Search 4: Adversarial Search ✓
- Part 3: Reasoning and Uncertainty
 - ① Reasoning 1: Constraint Satisfaction ✓
 - ② Reasoning 2: Logic and Inference ✓
 - ③ Probabilistic Reasoning 1: BNs ✓
 - ④ Probabilistic Reasoning 2: HMMs ✓
- Part 4: Planning
 - ① Planning 1: Intro and Formalism ✓
 - ② Planning 2: Algorithms & Heuristics ✓
 - ③ Planning 3: Hierarchical Planning ✓
 - ④ Planning 4: Stochastic Planning ✓
- Part 5: Learning
 - ① Learning 1: Intro to ML
 - ② Learning 2: Regression
 - ③ Learning 3: Neural Networks
 - ④ Learning 4: Reinforcement Learning
- Part 6: Conclusion
 - ① Ethical Issues in AI
 - ② Conclusions and Discussion

Week 2 Topic: Planning

- Planning 1: Intro and Formalism
- Planning 2: Algorithms & Heuristics
- Planning 3: Hierarchical Planning
- Planning 4: Stochastic Planning



Outline

1 Introduction and Formalism

► Introduction and Formalism

► Algorithms and Heuristics

► Hierarchical Planning

► Stochastic Planning

Introduction and Formalism

- Search versus Planning
- Planning Problem Representation
 - STRIPS
 - PDDL
- Forward and Backward Search



Outline

2 Algorithms and Heuristics

► Introduction and Formalism

► Algorithms and Heuristics

► Hierarchical Planning

► Stochastic Planning

Algorithms and Heuristics

- Planning Algorithms
 - Planning Graphs
 - Planning based on Heuristic Search
 - Satisfiability Testing



Outline

3 Hierarchical Planning

► Introduction and Formalism

► Algorithms and Heuristics

► Hierarchical Planning

► Stochastic Planning

Hierarchical Planning

- Control Knowledge in Planning
- Hierarchical Planning



Outline

4 Stochastic Planning

- ▶ Introduction and Formalism
- ▶ Algorithms and Heuristics
- ▶ Hierarchical Planning
- ▶ Stochastic Planning

Stochastic Planning

- Planning with Utilities
- Stochastic Planning Formalisms
- Dynamic Programming Algorithms for Stochastic Planning

Any Questions.